

The eukaryotic gut virome in hematopoietic stem cell transplantation: new clues in enteric graft-versus-host disease

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Much attention has been focused on the role of the bacterial microbiome in human health, but the virome is understudied. Although previously investigated in individuals with inflammatory bowel diseases or solid-organ transplants^{1,2}, virome dynamics in allogeneic hematopoietic stem cell transplantation (HSCT) and enteric graft-versus-host disease (GVHD) remain unexplored. Here we characterize the longitudinal gut virome in 44 recipients of HSCT using metagenomics. A viral ‘bloom’ was identified, and significant increases were demonstrated in the overall proportion of vertebrate viral sequences following transplantation ($P = 0.02$). Increases in both the rates of detection ($P < 0.0001$) and number of sequences ($P = 0.047$) of persistent DNA viruses (anelloriviruses, herpesviruses, papillomaviruses and polyomaviruses) over time were observed in individuals with enteric GVHD relative to those without, a finding accompanied by a reduced phage richness ($P = 0.01$). Picobirnaviruses were detected in 18 individuals (40.9%), more frequently before or within a week after transplant than at later time points ($P = 0.008$). In a time-dependent Cox proportional-hazards model, picobirnaviruses were predictive of the occurrence of severe enteric GVHD (hazard ratio, 2.66; 95% confidence interval (CI) = 1.46–4.86; $P = 0.001$), and correlated with higher fecal levels of two GVHD severity markers, calprotectin and $\alpha 1$ -antitrypsin. These results reveal a progressive expansion of vertebrate viral infections over time following HSCT, and they suggest an unexpected association of picobirnaviruses with early post-transplant GVHD.

Viruses, arising mainly from the Herpesviridae and Adenoviridae families, have been reported to be frequently linked to enteric GVHD^{3–5}. These viruses are associated with latent infections that can

be reactivated in the setting of host immunosuppression from HSCT. Active viral replication from reactivation, resulting in dynamic shifts in gut viral loads, may trigger intestinal inflammation and dysregulation of the restored immune response, which then leads to tissue damage and enhanced risk of GVHD. HSCT induces profound changes in bacterial microbiota diversity and may potentially impact the gut virome in a parallel fashion⁶.

Eukaryotic viruses are typically minor components of the gut as compared to phage populations⁷, but acute infections with exogenous viruses or chronic infections resulting from the reactivation of persistent viruses can give rise to high titers in individuals who are immunocompromised^{3,8}. Dynamic changes in the composition and diversity of viral populations following transplantation may have clinical significance, such as a recently described association between persistent viral infections and protective immunity during gut inflammation⁹. Broad surveillance approaches are needed to capture these changes, given the high degree of viral-sequence diversity and continual discovery of novel viruses in human stool by metagenomics^{10–12}.

In this study, we characterized the profile and longitudinal evolution of the human eukaryotic gut virome in individuals before and after HSCT to investigate potential links between changing viral dynamics and the development of enteric GVHD (Fig. 1). Here we focused on serial collection and analysis of stool samples, because the detection of human viruses in feces suggests active viral replication and shedding in the gut. In parallel, the bacterial microbiome and phage virome were also analyzed.

A total of 201 stool samples were collected from 44 individuals (Fig. 1), with a median of four samples per patient (range: 3–8 samples) (Supplementary Fig. 1). 31 individuals (70.5%) developed acute GVHD, 26 (83.9%) of whom experienced enteric GVHD, 16 of which (46.2%) were severe cases (stage 2 or higher) (Fig. 1a and Supplementary Table 1). For each week, the numbers of collected

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stool samples from patients with enteric GVHD were comparable (Fig. 1b), although proportionally fewer samples from individuals without enteric GVHD (heretofore defined as both patients without GVHD and those few with GVHD of another type) were available during later weeks (weeks 3–5). Acute GVHD—of any type—developed a median of 27.7 d after HSCT (interquartile range (IQR), 14–38 d) (Supplementary Table 1).

Metagenomic sequencing analysis using a virome-specific library preparation (Fig. 1c and Supplementary Results) yielded a total of 5.9 billion raw sequence reads, with a median of 29 million reads per sample (IQR, 25 million–33 million). Reads were analyzed using the SURPI computational pipeline for pathogen identification¹³, which detects and taxonomically classifies microbial reads by nucleotide (NT) alignment with those in the National Center for Biotechnology Information (NCBI) NT database, including the entirety of GenBank.

Parallel stool-microbiome analysis by sequencing of the V3–V4 region of the bacterial 16S ribosomal RNA (rRNA) gene showed that the most significant changes in relative abundance of bacterial taxa were an increase in *Lactobacillales* spp. ($P = 0.007$) and a concomitant decrease in *Clostridiales* spp. ($P = 0.02$) over time in individuals with enteric GVHD (Fig. 2a and Supplementary Results). Overall, there was no difference in richness ($P = 0.53$), but a slightly higher relative abundance ($P = 0.047$), of bacterial taxa at the genus level in patients with, relative to those without, enteric GVHD (Fig. 2b,c). Phage analysis of the metagenomic sequencing data showed that individuals with enteric GVHD had both decreased richness ($P = 0.01$) and a higher abundance of *Microviridae* prior to and during the first few weeks following transplantation ($P = 0.03$) (Fig. 2).

By SURPI analysis, the median percentage of viral reads overall was 0.22% (IQR, 0.001–1.6%). Eukaryotic viral reads accounted for a median of 1.76% of the viral fraction, although the proportion differed between individuals (IQR, 0.16–21.3%; range, 0–99.9%). Eukaryotic viruses accounted for more than 10% of total viral reads in 32.6% of stool samples, and more than 50% of total viral reads in 15.4% of stool samples. Eukaryotic viruses were assigned mostly to vertebrate taxa (86.9%), and plant viruses (12.1%) were the second most common (Supplementary Results).

Among vertebrate viruses, a total of 20 families, 52 genera and 347 species were identified. The most frequent viral families observed in stool samples from recipients of HSCT, in order of decreasing frequency, were Anelloviridae (37.8% of samples), Polyomaviridae (14.9%), Picobirnaviridae (13.9%) and Herpesviridae (10.9%) (Supplementary Fig. 2 and Supplementary Tables 2–4). Persistent DNA viruses (Anelloviridae, Herpesviridae, Papillomaviridae and Polyomaviridae) were detected in 100 of 201 samples (49.8%), whereas exogenous viruses were detected in 72 samples (35.8%). Repeat detection of the same viral family at least twice following transplantation was observed in 14 of 44 individuals (31.8%) harboring persistent viruses, four patients (9.1%) harboring exogenous viruses and nine (18.2%) harboring both types of virus (Supplementary Fig. 3). In general, persistent viruses were detected more frequently and yielded higher read numbers than exogenous viruses.

The overall proportion of vertebrate viral sequences in the guts of all HSCT recipients increased progressively during the weeks following transplantation ($P = 0.02$ for patients with or without enteric GVHD combined) (Fig. 3a). No difference in vertebrate viral diversity was observed between individuals with or without enteric GVHD ($P = 0.98$), but decreased richness at the species level was observed during the first few weeks following transplantation

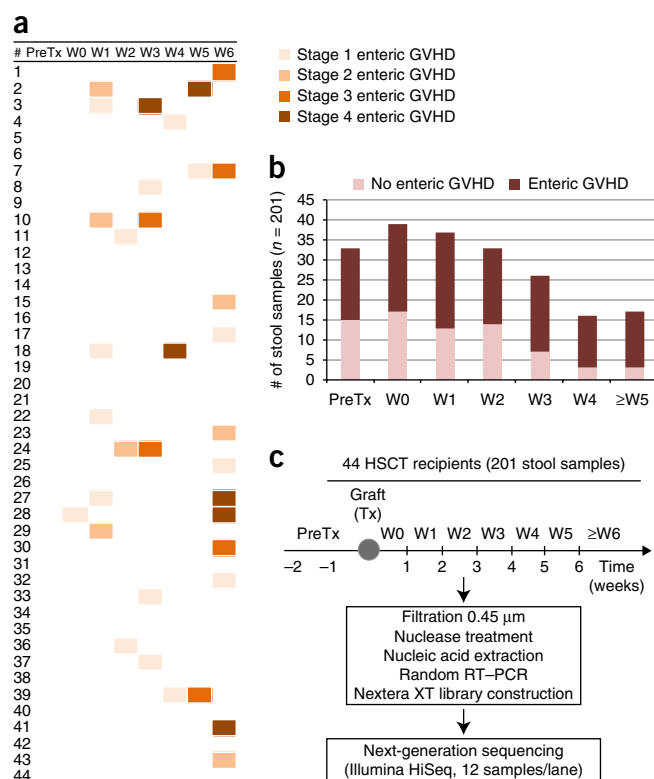


Figure 1 Collection of stool samples for virome analysis according to stage of enteric GVHD. A total of 201 samples were collected from 44 individuals. (a) For each patient (row), the stage of enteric GVHD according to week following HSCT is indicated by the color gradient. (b) Distribution of stool samples according to the week of collection ('W') and enteric GVHD status. (c) Flow chart of sample processing. Stool samples collected before HSCT ('PreTx') and every week after transplantation were transferred to the microbiology laboratory within 4 h and were then frozen. Samples were diluted in phosphate-buffered saline, filtered, treated with nuclease and extracted. Nextera XT libraries were obtained after performance of reverse transcription and random RT-PCR amplification, which was followed by sequencing on an Illumina HiSeq system. PreTx, the period before transplantation; Tx, transplant; HSCT, hematopoietic stem cell transplant; RT-PCR, reverse transcription–polymerase chain reaction.

in individuals with enteric GVHD ($P = 0.007$) (Fig. 3b,c). Notably, increases in both the rates of detection (Fig. 3d and Supplementary Table 5; $P = 0.047$) and the number of reads (Supplementary Fig. 4; $P < 0.0001$) of persistent DNA viruses were found in patients with enteric GVHD. More specifically, the number of Herpesviridae reads increased significantly in individuals with enteric GVHD relative to those without ($P = 0.009$) (Supplementary Fig. 4). By contrast, whereas RNA viral reads from picobirnaviruses (PBVs) were also more common in stool samples from individuals with enteric GVHD (Fig. 4a; 23.3% relative to 11.1%; $P = 0.039$), the proportion of PBV-positive samples decreased significantly in these patients over time and after week 2 post-transplant (Fig. 4b; 21 of 71 (29.6%) relative to 18 of 130 (13.8%); $P = 0.0008$).

Given the unexpected high frequency of PBV detection and known sequence divergence of PBV genomes, we reanalyzed all PBV reads (Supplementary Results) and confirmed their detection using targeted RT-PCR and Sanger sequencing (Supplementary Table 6). In total, PBVs were detected in 18 individuals, 12 (68.4%) of whom had PBV reads in multiple longitudinally collected samples (Supplementary Table 7). Most of the PBV-positive individuals had

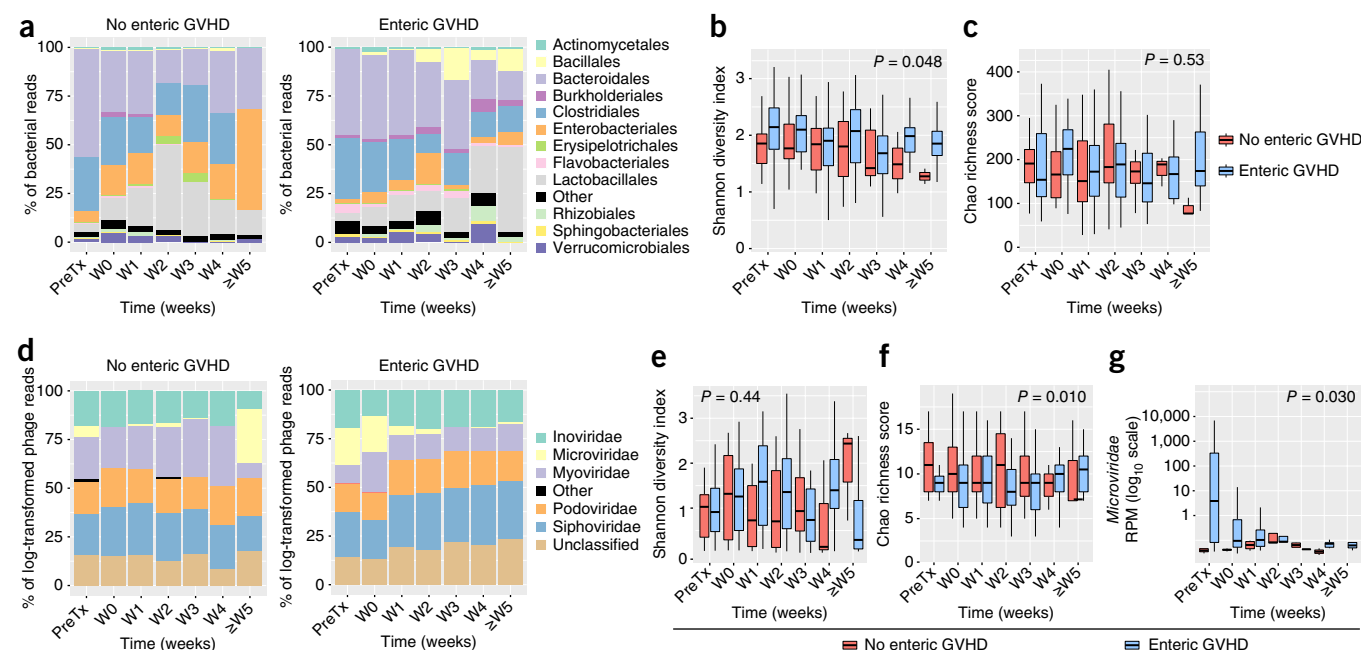


Figure 2 Microbiome and phage virome dynamics. **(a)** Relative proportion of sequence reads at the order level assigned to different bacterial taxa. **(b)** Bacterial alpha diversity, as measured at the genus level using the Shannon Diversity Index ($P = 0.048$, linear random-effects model and Wald test for joint significance of regression coefficients), by collection week in patients with or without enteric GVHD. **(c)** Richness, as measured at the genus level using the Chao Richness Score ($P = 0.010$, linear random-effects model and Wald test). **(d)** Relative proportion of reads at the family level assigned to phage taxa. **(e,f)** Phage virome alpha diversity (**e**, $P = 0.44$, linear random-effects model and Wald test) and richness (**f**, $P = 0.010$, linear random-effects model and Wald test), as measured at the genus level. **(g)** Microviridae counts in reads per million (RPM) over time in patients with and without enteric GVHD ($P = 0.030$, time-dependent log-transformed linear model and Wald test). In the box plots, the central rectangle spans the first quartile to the third quartile (the interquartile range or IQR), the central line inside the rectangle shows the median, and whiskers above and below the box show the locations of the minimum and maximum within 1.5 IQR of the lower quartile and the upper quartile, respectively.

been infected with strains related to those previously identified in humans or wastewater (Fig. 4 and Supplementary Table 8).

Among patient characteristics, including sex, age, type of graft, total body irradiation and human leukocyte antigen (HLA) matching, no notable associations were detected with GVHD of any type, enteric or nonenteric (Supplementary Table 1), with the sole exception of older age with severe enteric GVHD ($P = 0.005$). In addition, herpesvirus reactivation was not predictive of subsequent development of enteric GVHD ($P = 0.68$ and $P = 0.84$ for CMV and HHV-6 reactivation, respectively), as analyzed using a time-dependent Cox proportional-hazards model, despite more frequent detection of cytomegalovirus (viral load, $>3.5 \log_{10}$ IU/ml) and human herpesvirus 6 (HHV-6) in blood from individuals with enteric GVHD ($P = 0.04$ and $P < 0.0001$, respectively). Similarly, the development of GVHD was not correlated with viral host type, except for viruses of plants, which were associated with a lower risk of GVHD of any type (Supplementary Table 9).

Among all eukaryotic viruses, only the detection of PBV was predictive of later development of GVHD, both enteric and of any type (HR = 1.75; 95% CI = 1.07–2.85; $P = 0.02$), with a stronger association for severe enteric GVHD of stage 2 or higher (HR = 2.66; 95% CI = 1.46–4.86; $P = 0.001$) (Supplementary Table 9). Notably, in all patients who eventually developed enteric GVHD, the detection of PBV occurred before disease onset. Individuals positive for PBV also had significantly higher fecal levels of calprotectin ($P = 0.005$) and $\alpha 1$ -antitrypsin ($P = 0.01$), two biomarkers associated with severity and response to corticosteroids in individuals with GVHD¹⁴ (Fig. 4c,d and Supplementary Tables 10 and 11).

Here we describe a remarkable expansion of vertebrate viruses in feces from HSCT recipients during the first 6 weeks following transplantation. In patients with enteric GVHD, this bloom manifests as a marked increase in persistent DNA viruses and a corresponding decrease in phage richness. Surprisingly, an unexpected association of PBV infection with subsequent development of GVHD was also uncovered. We also report, consistent with a previous report of bacterial dynamics in mouse models of GVHD and individuals with GVHD-associated intestinal inflammation caused by GVHD⁶, an expansion of Lactobacillales and loss of Clostridiales in individuals with enteric GVHD⁶. Similarly, the finding of decreasing DNA phage reads from Microviridae over time in patients with GVHD was observed in a prior study interrogating stool samples from individuals with inflammatory bowel disease (IBD)².

Persistent DNA viruses in the Anelloviridae, Polyomaviridae and Herpesviridae families are among those most frequently detected in immunocompromised HSCT recipients, a development thought to be secondary to viral reactivation¹. Here we found that a larger proportion of patients with enteric GVHD than those without the condition had reactivation of DNA viruses in stool, and at higher levels. HHV-6 and CMV reactivation in the blood ($>3.5 \log_{10}$ UI/ml) were also more frequent in individuals with enteric GVHD than in those without ($P < 0.01$ and $P = 0.049$, respectively). Nevertheless, reactivation of persistent DNA viruses, whether in stool or blood, was not predictive of GVHD (enteric or any type). Rather, viral reactivation occurred at later time points following transplantation (weeks 3–5) in patients developing enteric GVHD, and thus might have been precipitated by GVHD-associated inflammation and/or immunosuppression associated

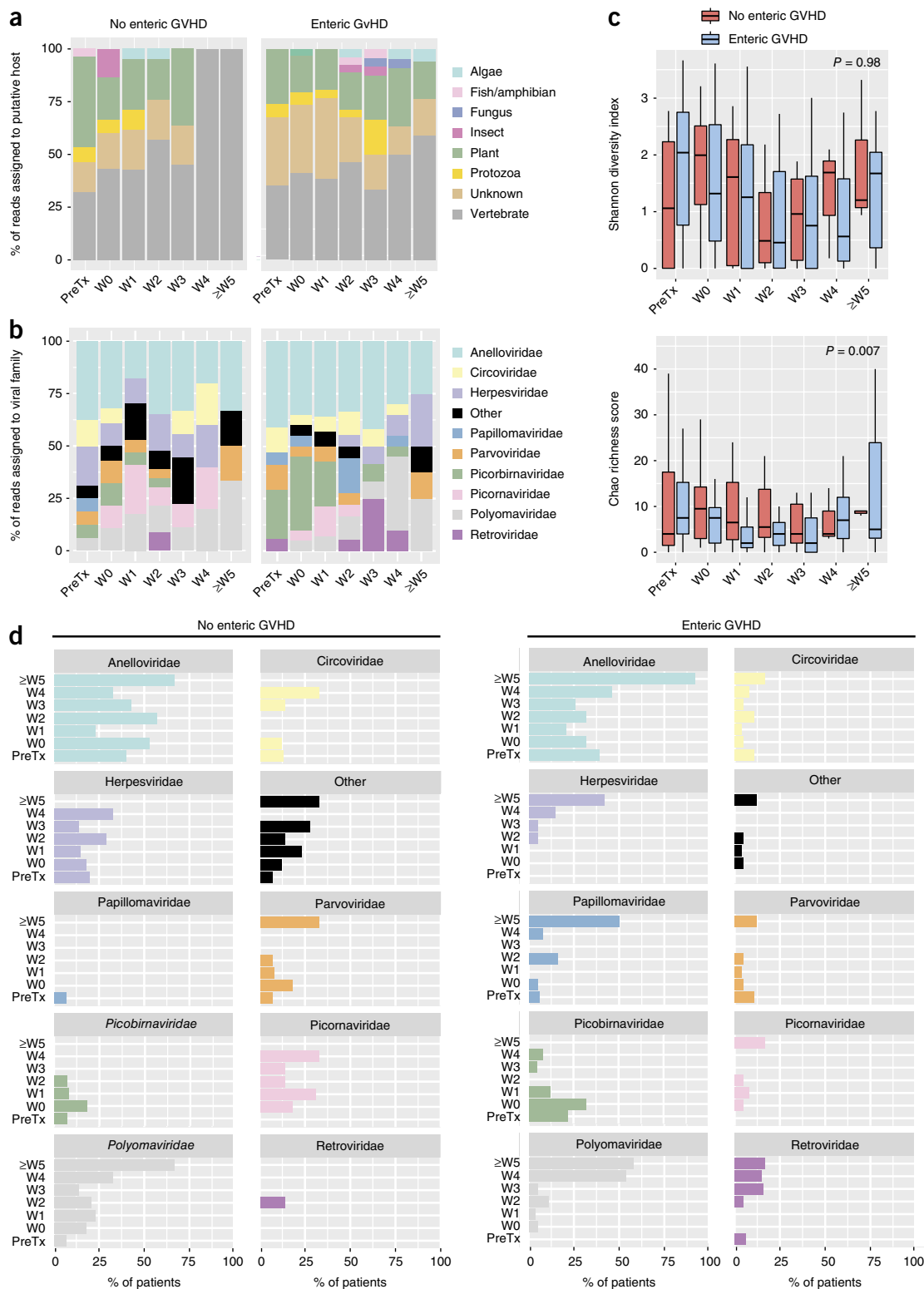


Figure 3 Evolution of the enteric virome following transplantation. **(a)** Distribution of viral reads across putative eukaryotic hosts. The percentage of reads corresponding to each host is shown for each time point. **(b)** Distribution of viral reads across different viral families. The percentage of viral reads corresponding to each family is shown for each time point. **(c)** Viral alpha diversity (Shannon Diversity Index, $P = 0.98$, linear random-effects model and Wald test for joint significance of regression coefficients) and richness (Chao Richness Score, $P = 0.007$, linear random-effects model and Wald test) of vertebrate viruses at the species level by week of collection in patients with or without enteric GVHD. **(d)** For each viral family, the proportion of patients positive for sequence reads from that family is plotted longitudinally over time. The color scheme is the same as in **b**. Summary data are stratified according to the presence or absence of enteric GVHD. A positive viral detection is defined as ≥ 15 reads, and viral families detected at a frequency lower than 5% are classified as “other”.

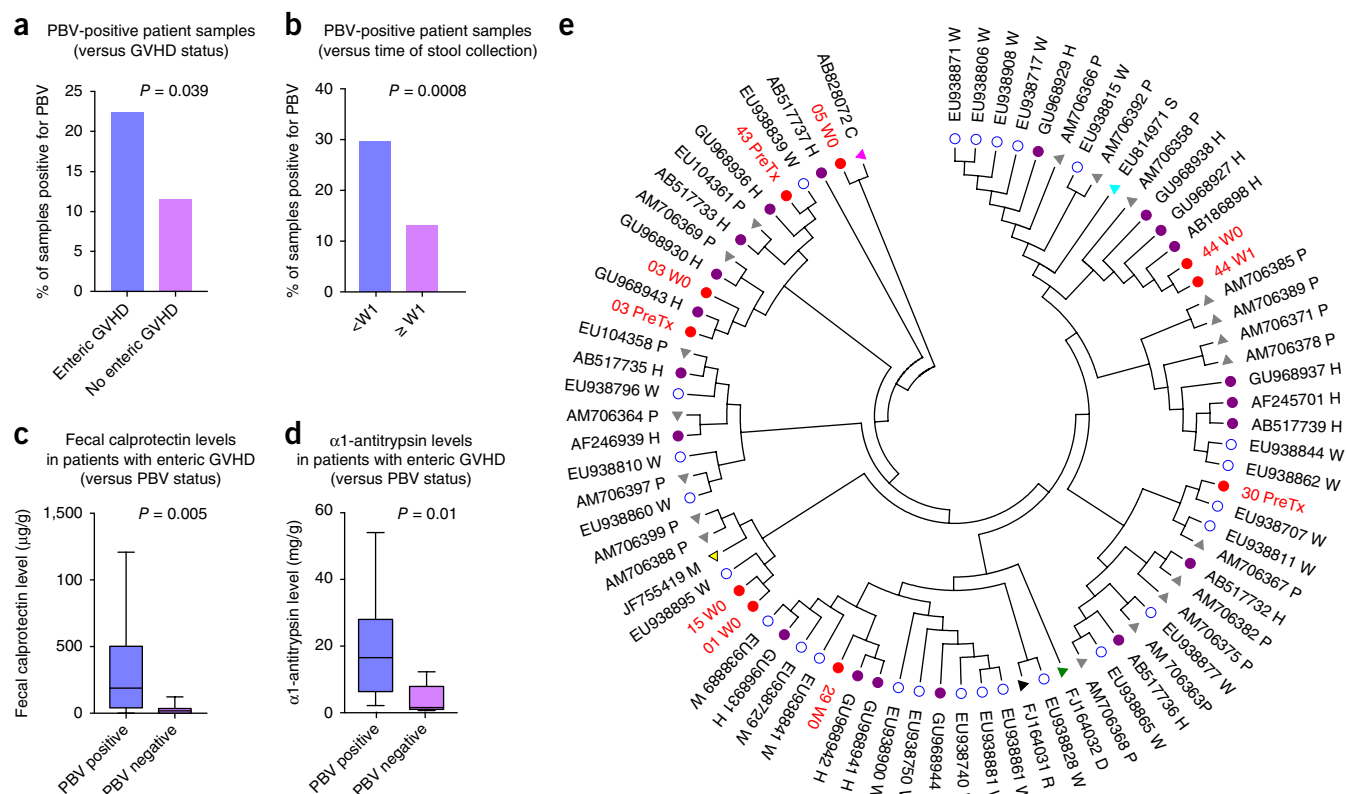


Figure 4 Frequency and diversity of PBVs in individuals with or without enteric GVHD. (a) Percentage of stool samples positive for PBV among those with or without enteric GVHD ($P = 0.039$, Fisher's exact test). (b) Frequency of detection of PBVs before transplantation or within the first week after transplant (<W1) or at a later time point (≥W1) in all patients ($P = 0.0008$, Fisher's exact test). (c) Fecal concentrations of calprotectin in individuals with enteric GVHD, with (mean = 330.5 μg/g) or without PBV infection (mean = 40.3 μg/g) ($P = 0.005$, Mann–Whitney U test). (d) Fecal concentrations of α1-antitrypsin in patients with enteric GVHD with (mean = 11.92 mg/g) or without PBV infection (mean = 4.43 mg/g) ($P = 0.01$, Mann–Whitney U test). (e) Phylogenetic tree of a partial amino acid (aa) sequence of the PBV RNA-dependent RNA polymerase (RdRP), corresponding to aa 208–263 of reference PBV strain AB186898 (genogroup 1), was generated using a maximum-likelihood method with Poisson correction²³. The reference sequences selected for the phylogeny included all PBV sequences available in the NCBI NT database with coverage of the corresponding RdRP partial sequence^{20,21,24–26}. Sequences were aligned using Geneious 9.0.2 (Biomatters, Auckland, New Zealand), and trees were built using Mega 6 (ref. 23). PBV sequences from the current study ($n = 12$) are highlighted with a filled red circle and in boldface red text. Sequences previously detected in human samples are designated by the letter H with a filled purple circle. The designations C (pink triangle), D (green triangle), M (yellow triangle), P (gray triangle), R (black triangle), S (cyan triangle) and W (blue open circle) stand for cow, dog, mouse, porcine, rat and snake hosts, or wastewater, respectively.

with corticosteroid therapy. These findings suggest that the onset of GVHD might have triggered DNA viral reactivation, rather than reactivation constituting a risk factor for GVHD, per se. Indeed, the reactivation of persistent DNA viruses in the gut during weeks 1–3 after transplantation was seen mainly in individuals without enteric GVHD. This suggests a potential protective role for persistent DNA viral replication in the regulation of gut inflammation predisposing to enteric GVHD, as previously postulated in a mouse model of IBD⁹.

Surprisingly, the presence of plant viruses was associated with a reduced risk of GVHD of any type. It is possible that HSCT recipients without GVHD have an increased likelihood of advancing to a more robust diet. Previous reports show that a specific diet modifies the gut microbiome⁷, and that these alterations in the gut microbiota can mitigate the severity of enteric GVHD^{15,16}. A parallel mechanism may be present with respect to the plant virome, given that dietary antigens, including those derived from viruses, are known to have a role in intestinal immunity¹⁷.

Notably, we found that evidence of stool PBV infection was predictive of later development of GVHD. Because the Cox proportional-hazards model used here takes into account all time points, we cannot pinpoint a precise time point corresponding to maximal risk, although the proportion of individuals with enteric GVHD found to be positive

for PBV (Fig. 3b) suggests that it might fall near the time of transplantation. Data on the pathogenicity of Picobirnaviridae remain limited and inconclusive¹⁸. PBV surveillance efforts have been hampered by the high genetic variability of the virus^{19–21}, limiting accurate rates of detection in clinical samples. Phylogenetic analyses showed that PBV sequences in HSCT recipients clustered broadly with other PBV sequences identified mostly in human or wastewater samples (Fig. 4). These results suggest that the detection of PBV represents bona fide human infections from these viruses. In addition, PBV strains from individual patients were very different from each other according to phylogenetic analysis, excluding inadvertent cross-contamination between samples as an explanation for the observed strain diversity.

Repeat detection of PBV was observed in longitudinally collected samples from individuals with GVHD. Although chronic excretion of PBV for several months has been previously described in patients with HIV infection experiencing diarrhea²², our finding that serial PBV sequences from the same patient are closely related underscores the capacity of individual PBV strains to cause persistent infection, which potentially promotes GVHD development by maintaining an inflammatory milieu in the gut. Whereas biomarkers associated with acute GVHD and GVHD severity have been described in the literature,

to our knowledge, there are no biomarkers currently being used clinically to predict GVHD. Further investigations and independent validation of the potential role of PBV infection in GVHD onset and its utility as a predictive biomarker are needed.

METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the [online version of the paper](#).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

J.L. conceived of the study, performed the experiments, analyzed the data, designed the figures and wrote the manuscript. M.R.-R. performed the statistical analyses and designed the figures. J.B. ran PCR confirmation for picobirnaviruses and analyzed the data. M.R. collected clinical data. S.M.-D., prepared patient samples for virome analysis and microbiome analysis. S.N.N., S.F. and E.S. contributed to the analysis of patient, microbiome and virome data. C.R. and N.K. contributed to the analysis of microbiome and stool biomarkers. P.P. contributed to statistical analysis. F.S. provided funding and resources and contributed to the discussion. G.S. conceived of the study, analyzed the data and wrote the manuscript. C.Y.C. conceived of the study, analyzed the data, designed the figures and wrote the manuscript. All authors provided feedback on the manuscript. F.S., G.S., J.L. and C.Y.C. supervised the project. J.L. and C.Y.C. secured funding for the study.

COMPETING FINANCIAL INTERESTS

The authors declare competing financial interests: details are available in the [online version of the paper](#).

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ONLINE METHODS

Ethics statement and inclusion criteria. Patients analyzed in this study were all hospitalized for allogeneic hematopoietic stem cell transplant in the Hematology and Transplantation unit of Saint-Louis hospital, Paris, France, and enrolled in the CryoStem cohort, sponsored by the French National Research Agency (ANR). Inclusion criteria were adults (18 years old or older) who underwent allogeneic hematopoietic stem cell transplantation and had at least three weekly stool samples collected over a period starting before transplantation (between the conditioning regimen and engraftment) and up to 8 weeks after engraftment (**Supplementary Fig. 1**). Our objectives were to include at least 40 participants. Between 1 April 2013 and 31 March 2014, 44 patients fulfilled inclusion criteria. All patients provided written informed consent for enrollment in the study. Approval was received from the research ethics committee of "Île de France IV" (IRB #00003835). The study, a non-interventional research study with no additional clinical procedures, was carried out in accordance with the Declaration of Helsinki. Data analyses were carried out using a database with all patient identifiers removed.

Patient cohort data. Clinical data extracted from patient medical records included age, gender, diagnosis, HLA matching between donor and recipient, conditioning regimen, T cell depletion, presence or absence of GVHD, and graft characteristics. In individuals with GVHD, the location, grade, chronicity and treatment were recorded. If relapse or death occurred, the dates and causes of death were also recorded. Clinical characteristics are detailed in **Supplementary Table 1**. All patients underwent gut decontamination and received the same antibiotic regimen (ofloxacin and amoxicillin) before HSCT.

Assessment of enteric graft-versus-host disease. Acute enteric GVHD was suspected if patients developed the following symptoms after transplantation: diarrhea, nausea, emesis or anorexia. In the absence of contraindications such as thrombocytopenia, ileus, suspicion of intestinal perforation or patients declining to participate, an endoscopy with histologic examinations was carried out. Acute GVHD was diagnosed and graded according to Glucksberg and Consensus Criteria^{27,28}. First-line treatment with corticosteroids at a dose of 1–2 mg/kg/d was initiated in all individuals with acute GVHD grade 2 or higher. Corticosteroid-refractory GVHD was defined either as progression at day 3, stable disease on day 7 or the absence of remission 2 weeks after the initiation of corticosteroids. Histological criteria of enteric GVHD were assessed from biopsies of the gastrointestinal tract, as described previously²⁹.

In stool samples, the concentrations of calprotectin and $\alpha 1$ -antitrypsin were determined in stool samples, as previously described³⁰. Briefly, fecal aliquots were frozen at -80°C to quantify calprotectin using a 'sandwich'-type enzyme-linked immunosorbent assay (ELISA) with polyclonal Ab (Calprest, Eurospital SPA, Trieste, Italy). Results for calprotectin were reported as μg per g wet weight of stools, with a measurement range of 10–5,000 $\mu\text{g/g}$ and 50 $\mu\text{g/g}$ stool set as the upper limit of normal. Fecal $\alpha 1$ -antitrypsin measurement was determined by immunonephelometry on the BN ProSpec system (Siemens SAS, Saint-Denis, France) and reported as mg per g dry weight of stools. The measurement range was between 0.01 mg/g and 20 mg/g dry weight, and the upper limit of normal was set at 1.5 mg/g dry weight.

Clinical diagnosis of enteric GVHD was further supported by fecal calprotectin levels higher than 50 $\mu\text{g/g}$ of stool, or levels of $\alpha 1$ -antitrypsin higher than >1.5 mg/g of stool (dried sample) or the observation of histological findings of enteric GVHD. In our series, fecal calprotectin testing was available for all 23 individuals with enteric GVHD, $\alpha 1$ -antitrypsin concentrations available for 16 patients and intestinal biopsies for 20 patients.

Sample collection and nucleic acid extraction. Investigators were blinded to patient GVHD disease status and clinical outcomes at the time of sample processing and next-generation sequencing analyses. Stool samples were collected a few days before HSCT and once a week after HSCT until week 8 or time of hospital discharge. Samples were sent to the microbiology laboratory at Hospital Saint-Louis and were frozen at -80°C within 4 h after collection. Stool specimens were diluted (20%) in phosphate-buffered saline. The suspension (1,500 μl) was centrifuged for 5 min at 10,000g. To enrich for viral

particles by reduction of host background, stool supernatant was filtered through a 0.45- μm filter (Corning Costar Spin-X centrifuge tube filters), and an aliquot of 315 μl of filtrate was pretreated before extraction by incubation with a cocktail of nucleases: (i) 20 μl of TURBO DNase (Invitrogen, Carlsbad, CA); (ii) 8.5 μl of Baseline-ZERO DNase (Ambion, Foster City, CA); (iii) 8.5 μl of Benzonase (Novagen); and (iv) 8.5 μl of RNase A (Roche Applied Biosciences, Meylan Cedex, France) for 30 min, at 37°C .

Our approach of performing viral metagenomics of filtered stool, pre-treated with nuclease to enrich for encapsidated DNA and RNA viruses, is well established in the literature, and rigorous comparison between the different methods has been described, with filtration and nuclease treatment appearing to cause only minor shifts in viral abundance^{31,32}. This is in contrast to the use of polyethylene glycol (PEG) precipitation and CsCl density centrifugation for viral purification, which does bias against phages and affects quantitative accuracy³¹.

Library preparation and viral metagenomic next-generation sequencing.

A random-amplification approach was used for virome analysis instead of multiple-displacement amplification, which has a detection bias for small circular DNA viruses such as phages. Nucleic acids were converted to complementary DNA (cDNA) using random nonamer primers attached to a linker sequence ('primer B') in a 30- μl reaction with Superscript III reverse transcriptase (Invitrogen), followed by second-strand using Sequenase (Affymetrix, Santa Clara, CA) and 25 cycles of PCR amplification using high-fidelity KlenTaq LA DNA polymerase (Sigma-Aldrich, Santa Clara, CA), as previously described^{13,33}. Ultra-pure bovine serum albumin (BSA) (Ambion) was added into reverse transcription and PCR steps to reduce potential PCR inhibition. In separate experiments, we verified that the BSA was not contaminated with any of the viruses detected in HSCT recipients and presented here (LeGoff, Mercier-Delarue and Simon, data not shown). Amplified cDNA was purified using AMPure XP beads (Beckman Coulter, Brea, CA) and quantitated using the Qubit Fluorometer and Qubit dsDNA HS Assay Kit (Life Technologies, Carlsbad, CA). 2 ng of dsDNA were used as input for the Nextera XT kit for NGS library generation. Nextera XT libraries were purified using an epMotion 5075 TMX BioRobot (Eppendorf, Hamburg, Germany). Libraries were sequenced on an Illumina HiSeq 2500 in rapid mode using 150/150-bp paired-end sequencing (12 stool samples per lane).

To minimize the risk of contamination, reagents were stored and prepared in a DNA-free room. Samples were treated in batches, with a maximum of 12 samples at a time, and were handled in a laminar flow hood on ice. During the whole procedure, tubes were opened individually just before transfer to the reaction tube and were closed before work began on the next sample. All surfaces were cleaned with 10% sodium hypochlorite before and after usage to remove any nucleic acids.

To test for the presence of potential contaminants in the reagents used for DNA extraction and sequencing library preparation and any cross-contamination between samples during the procedure, we sequenced four negative controls in parallel, one for each batch of samples. In total, we found between 0 and 23 reads from eukaryotic viruses (Cyprinid herpesvirus 3, *Plutella xylostella* multiple nucleopolyhedrovirus, Prunus necrotic ringspot virus and Equid herpesvirus 1), with a maximum of 14 reads of Equid herpesvirus 1 for one sample. Of note, the herpesvirus sequences were found to correspond to vector-sequence contamination or low-complexity sequences on manual examination. No reads related to the viruses that were found in clinical samples and presented in the main text were detected in the negative control samples. To further minimize the risk of misinterpretation from potential sample cross-contamination, we used 15 as the threshold cutoff value for number of reads to define a sample as positive for a virus.

Bioinformatics analysis. Next-generation sequencing (NGS) data sets were analyzed for pathogens using SURPI, a bioinformatics pipeline for pathogen detection and discovery from metagenomic sequencing data¹³. The SURPI pipeline first preprocesses the raw data, with trimming of adapters and removal of low-quality and low-complexity sequences. Human host sequences are then identified and computationally subtracted, with the remaining reads mapped to the NCBI reference databases for comprehensive identification of viruses, bacteria, fungi and parasites. For viral identification, the SNAP nucleotide

aligner³⁴ is run using an edit distance of 12 against the NCBI NT database (June 2013), which includes GenBank, enabling the detection of reads with $\geq 90\%$ identity to reference sequences in the database. Next, the translated nucleotide aligner RAPSearch³⁵ is used to detect divergent viral reads by translated nucleotide alignment to the NCBI nonredundant (NR) protein database. For this study, the RAPSearch results were used only for the identification of divergent PBV sequences. Viral reads were filtered to exclude misalignments by BLASTn analysis of a subset of sequences assigned to unique GenBank identifier (GI) numbers at an E-value threshold cutoff of 10^{-8} , followed by taxonomic classification to the appropriate rank (family, genus or species) using a lowest-common-ancestor algorithm incorporating the SNAP nucleotide aligner³⁴. A sample was considered to be positive for a virus family if the number of reads matching to that family was at least 15. For quantitative comparison between samples, reads were normalized according to the number of pre-processed reads (adaptor-trimmed reads filtered to exclude low-quality and low-complexity sequences) and expressed in RPM.

Nucleic acid extraction and sequencing for microbiome analysis. Bacterial 16S microbiome libraries were prepared from total DNA extracted using the Stool DNA Isolation Kit (Norgen Biotek Corp, Thorold, ON, Canada). Variable regions V3–V4 were targeted using a proprietary method (Norgen Biotek), and Nextera XT sample libraries were prepared (Illumina). Samples were sequenced at a read length of 300 bp on the Illumina Miseq platform (96 samples per run). Sequence analysis was performed on the instrument using the Illumina 16S metagenomic workflow, a high-performance implementation of the Ribosomal Database Project Classifier³⁶ on an Illumina-curated version of the Greengenes taxonomic database (last updated May 2013).

Picobirnavirus PCR amplification. To assess the specificity of PBV detection by NGS and SURPI identification, we performed viral RT–PCR followed by confirmatory Sanger sequencing. Attempts to test short conserved primer sets designed *de novo* or previously published degenerate primer sets for detection of PBV failed owing to high intersample viral-sequence diversity (J. LeGoff, J. Bouquet, S. Mercier-Delarue, F. Simon, G. Socié & C.Y. Chiu, unpublished data). Thus, we designed primers on the basis of the recovered PBV reads for each sample to be used in individual RT–PCR reactions. All PBV reads from each sample were aligned by BLASTn to all known PBV nucleotide sequences in NCBI Genbank NT at an E-value cutoff of 1×10^{-7} and by BLASTx to all known PBV proteins in at an E-value cutoff of 1×10^{-5} . Batchprimer3 (ref. 37) was used to design primers derived from the PBV read with the best match to PBV reference sequences in NCBI GenBank and targeting a 100-bp amplicon size and melting temperature of 55 °C (Supplementary Table 6). PBV was amplified using the OneStep RT–PCR kit (Qiagen) with 40 cycles of PCR and the addition of 1% BSA, according to the manufacturer's instructions. PCR amplicons were visualized by 2% agarose gel electrophoresis. Bands of the expected size were cloned into a TOPO-TA plasmid vector (Life Technologies), transformed into Mix & Go *Escherichia coli* competent cells strain 5a (Zymo Research), and Sanger sequenced using M13 primers flanking the amplicon inserts.

Quantification of cytomegalovirus, Epstein–Barr virus (EBV) and human herpesvirus 6. CMV and EBV were detected and quantified on the Abbott m2000 platform with the Abbott real-time CMV assay and the EBV PCR Kit V1 assay (Abbott Molecular, Des Plaines, IL), using 500 μ l of whole blood for each. For the detection and quantification of HHV-6, nucleic acids were first extracted from 200 μ l of whole blood by using the QIA Symphony DSP DNA Mini Kit on the QIA Symphony instrument (Qiagen) and then amplified with the RealStar HHV-6 PCR Kit 1.0 (Altona Diagnostics) on an ABI 7500 system.

CMV loads were expressed in international units per ml (IU/ml) and EBV and HHV-6 loads in copies/ml. Limits of detection for CMV, EBV and HHV-6 were 62 IU/ml, 1,000 copies/ml and 800 copies/ml, respectively.

Statistics. Proportions were compared using Fisher's exact test. The variation in numbers of reads according to weeks was evaluated using a time-dependent linear model after log transformation of the number of reads. The association of detected viruses with risk of GVHD (enteric or nonenteric) was assessed using a Cox proportional-hazards model for time-dependent covariates. Results were presented as hazard ratios at a 95% confidence interval, and *P* values corresponding to the statistical-significance scores. Associations between continuous measurement and GVHD were assessed using a linear random-effects model (according to weeks). Associations between binary measurement and GVHD were assessed using a logistic random-effects model (according to weeks). The Shannon Diversity Index (otherwise known as the Shannon–Weaver or Shannon–Wiener Diversity Index) was calculated as $H = -\sum (p_i \log(p_i))$, where p_i is the proportional abundance of species *i* and calculation is summed across all species. The Chao Richness Score was calculated as the number of different species. All measurements were obtained using R package *vegan*2.4-2 (<http://vegan.r-forge.r-project.org/>, accessed June 2017). All statistical tests were calculated as two-sided at the 0.05 significance level. Analyses were performed using the R statistical package (<http://www.R-project.org>, accessed June 2017). In box plots, the central rectangle spans the first quartile to the third quartile (the interquartile range or IQR), the central line inside the rectangle shows the median, and whiskers above and below the box show the locations of the minimum and maximum within 1.5 IQR of the lower quartile and the upper quartile, respectively.

Data availability. Reads with human sequences removed by Bowtie2 (ref. 38) high-sensitivity local alignment to the human genome (GRCh38/hg38 build) have been deposited in the NCBI Sequence Read Archive (SRA). Additional detailed information on experimental design and reagents for this study is provided in the **Life Sciences Reporting Summary**.

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Life Sciences Reporting Summary

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Experimental design

1. Sample size

Describe how sample size was determined.

No sample size was calculated. Patients analyzed in this study were hospitalized for allogeneic hematopoietic stem cell transplant. Inclusion criteria were adult patients (18 years old or older) who underwent allogeneic hematopoietic stem cell transplantation and had collection of at least 3 weekly stool samples collected over a period starting before transplantation (between the conditioning regimen and engraftment) and up to 8 weeks after engraftment. Our objectives were to include at least 40 patients. Between April 1st 2013 and March 31 2014, the 44 patients who fulfilled inclusion criteria were consecutively included in the study without any selection.

2. Data exclusions

Describe any data exclusions.

No data were excluded.

3. Replication

Describe whether the experimental findings were reliably reproduced.

Not applicable

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

All patients were treated by allogeneic hematopoietic stem cell transplant. There was no randomization. All patients fulfilling inclusion criteria were consecutively included and evaluated.

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

Investigators and technicians were blinded to patient graft-versus-host disease status and clinical outcomes at the time of sample processing and next-generation sequencing analyses.

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
- ☐ ☒ A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ A statement indicating how many times each experiment was replicated
- ☐ ☒ The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section)
- ☐ ☒ A description of any assumptions or corrections, such as an adjustment for multiple comparisons
- ☐ ☒ The test results (e.g. P values) given as exact values whenever possible and with confidence intervals noted
- ☐ ☒ A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
- ☐ ☒ Clearly defined error bars

See the web collection on [statistics for biologists](#) for further resources and guidance.

Software

Policy information about [availability of computer code](#)

7. Software

Describe the software used to analyze the data in this study.

R package vegan 2.4-2; R statistical package (version 3.3.2. online at <http://www.R-project.org>)

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* [guidance for providing algorithms and software for publication](#) provides further information on this topic.

Materials and reagents

Policy information about [availability of materials](#)

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.

Materials are human blood and stool samples. Their availability is limited.

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).

No antibodies used in this study

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.

No eukaryotic cell line used in the study.

b. Describe the method of cell line authentication used.

Not applicable

c. Report whether the cell lines were tested for mycoplasma contamination.

Not applicable

d. If any of the cell lines used are listed in the database of commonly misidentified cell lines maintained by [ICLAC](#), provide a scientific rationale for their use.

Not applicable

Animals and human research participants

Policy information about [studies involving animals](#); when reporting animal research, follow the [ARRIVE guidelines](#)

11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.

Not applicable

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.

Patients analyzed in this study were all hospitalized for allogeneic hematopoietic stem cell transplant in the Hematology and Transplantation unit of Saint-Louis hospital, Paris, France. Inclusion criteria were adult patients (18 years old or older) who underwent allogeneic hematopoietic stem cell transplantation and had collection of at least 3 weekly stool samples collected over a period starting before transplantation (between the conditioning regimen and engraftment) and up to 8 weeks after engraftment. Clinical data extracted from patient medical records included age, gender, diagnosis, HLA matching between donor and recipient, conditioning regimen, T-cell depletion, presence or absence of GVHD, and graft characteristics. In patients with GVHD, the location, grade, chronicity, and treatment were recorded. If relapse or death occurred, the dates and cause of death were also recorded.